

Update Joint Modeling 07/09

Overview Methods:

The three suggested methods are implemented

1. Classical ML
2. MI + closed form
 1. imputations were done within clusters, 3 imputations per clusters, with a maximum of 10 repetitions.
3. Closed form + iterative reweighting:
 1. **Currently reweighting until either max Δ of β estimates $< 1e-6$ or 30 iterations reached. Doing simple iterative reweighting: improvements are possible.**
 2. **For the closed form, each cluster needs at least 3 observations in the current model (random intercept + random slope), current solution: exclude all clusters with too little observations. Perhaps better to only simulate data with at least**

Simulation setup

Formula:

$$y_{ij} = \beta_0 + \beta_1 Treatment_i + \beta_2 Time_j + \beta_3 Treatment_i * Time_j + b_{0i} + b_{1i} * Time_j$$

Parameters were taken from the toenail data:

Effect	Parameter	Size
Intercept	β_0	2.4562
Treatment Effect	β_1	0 (<i>0.593 in toenail data (this was a dose effect)</i>)
Time Effect	β_2	0.2792
Treatment x Time	β_3	0.0350
Random intercept variance	d_{11}	7.3174
Random slope variance	d_{22}	0.2239
Random effects covariance	d_{12}	-0.4985
Residual variance	σ^2	3.1508

Variable	Value
number of clusters	100, 50, 20, 10 (<i>10.000 clusters was asked too, but currently excluded due to large computing time</i>)
number of repeated measures	12
missingness pattern	“half-missing” (half of the clusters have no missingness, other half have uniform probability to have 1-11 observations, dropout is monotone)
number of simulations / scenario	10.000

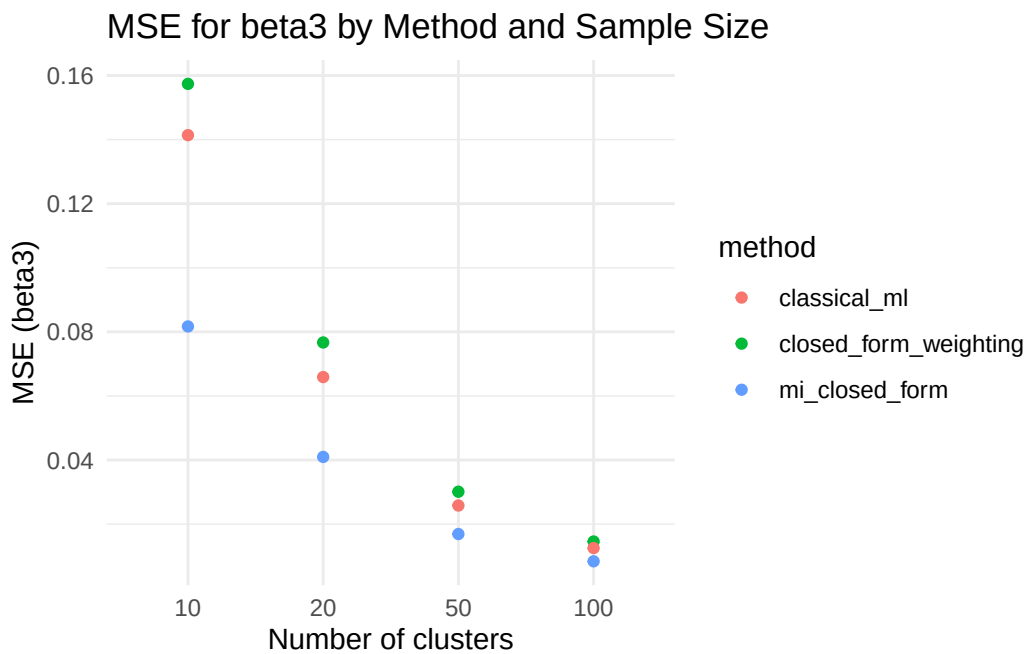
Next steps

1. Improve convergence on reweighting
2. Implement Type I error rate
3. Joint model

Results

Mean Squared Error

	method	n	mse_beta0	mse_beta1	mse_beta2	mse_beta3
9	classical_ml	100	0.1747600	0.3434405	0.006287665	0.012526373
1	closed_form_weighting	100	0.1923331	0.3825597	0.007264226	0.014573228
5	mi_closed_form	100	0.1733741	0.3333833	0.005462391	0.008398979
10	classical_ml	50	0.3400117	0.6916183	0.012792069	0.025812234
2	closed_form_weighting	50	0.3793336	0.7718828	0.014936174	0.030109505
6	mi_closed_form	50	0.3370610	0.6708683	0.010986122	0.016903518
11	classical_ml	20	0.8552565	1.7253949	0.032641477	0.065899944
3	closed_form_weighting	20	0.9518171	1.9291590	0.037682439	0.076691408
7	mi_closed_form	20	0.8504099	1.6788060	0.027075826	0.040969445
12	classical_ml	10	1.7496850	3.4388541	0.070083670	0.141382976
4	closed_form_weighting	10	1.9652033	3.8626932	0.078508759	0.157388881
8	mi_closed_form	10	1.7292583	3.3605714	0.053502513	0.081690039



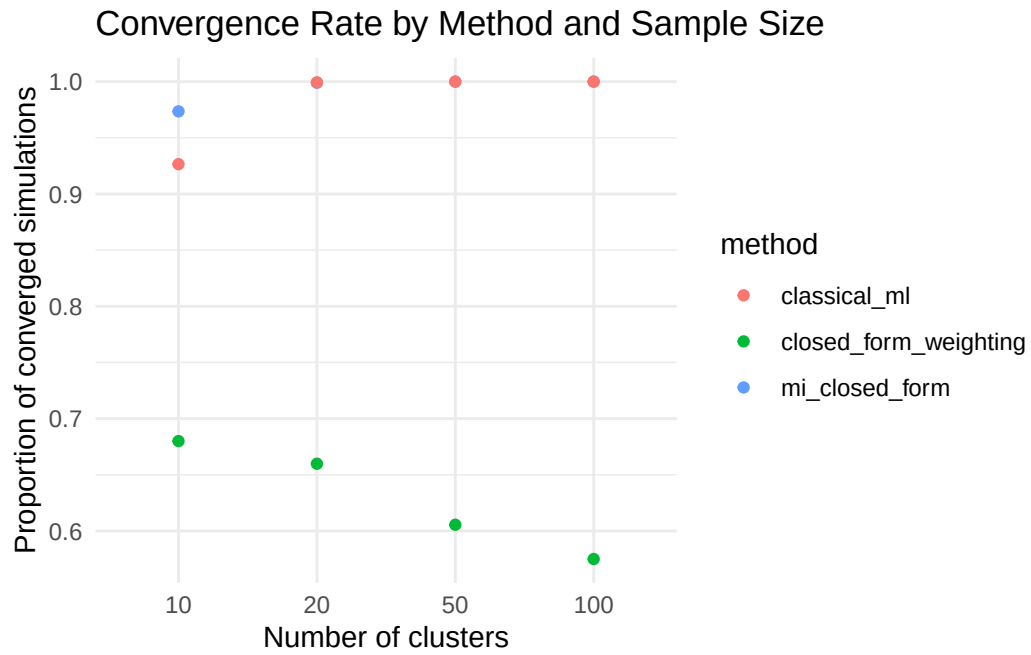
Mean Absolute Bias

	method	n	mean_abs_bias_beta0	mean_abs_bias_beta1
9	classical_ml	100	0.3351289	0.4682317

1	closed_form_weighting	100	0.3512660	0.4948424
5	mi_closed_form	100	0.3330793	0.4611698
10	classical_ml	50	0.4640112	0.6622293
2	closed_form_weighting	50	0.4905630	0.7004840
6	mi_closed_form	50	0.4627778	0.6522791
11	classical_ml	20	0.7396310	1.0433010
3	closed_form_weighting	20	0.7776917	1.1002045
7	mi_closed_form	20	0.7371647	1.0302838
12	classical_ml	10	1.0562568	1.4870427
4	closed_form_weighting	10	1.1177040	1.5721710
8	mi_closed_form	10	1.0498128	1.4678727
	mean_abs_bias_beta2	mean_abs_bias_beta3		
9	0.06303009	0.08909957		
1	0.06734696	0.09629941		
5	0.05868202	0.07281778		
10	0.08992678	0.12845421		
2	0.09729618	0.13860277		
6	0.08339874	0.10359881		
11	0.14360628	0.20485811		
3	0.15422608	0.22090788		
7	0.13128094	0.16076191		
12	0.21040152	0.29948711		
4	0.22184554	0.31588889		
8	0.18466193	0.22392158		

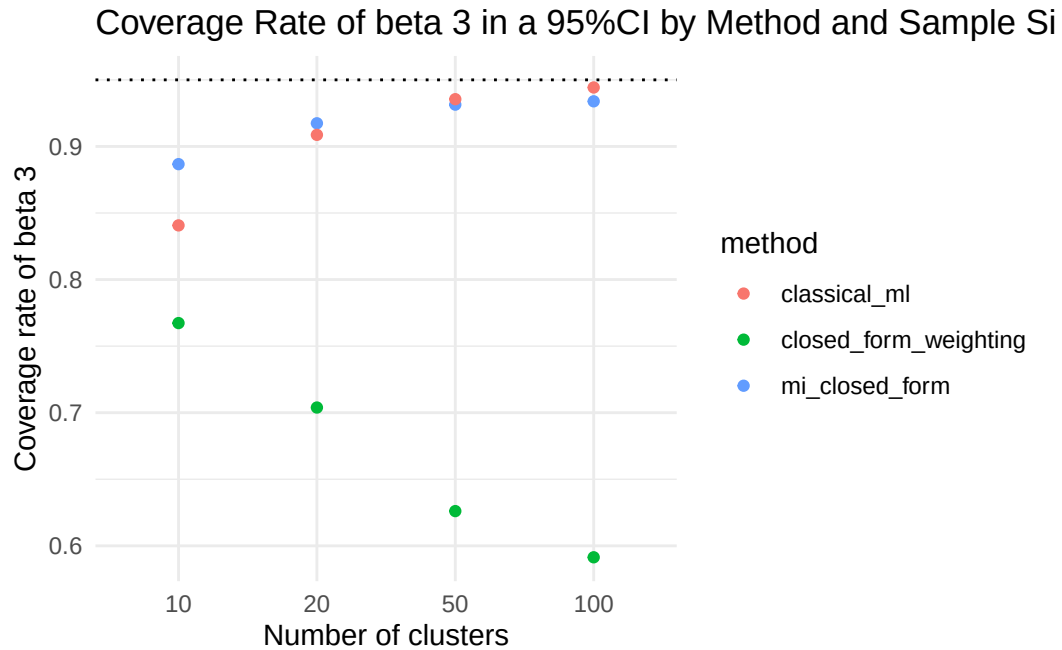
Convergence Rate

	method	n	prop_converged_ok	prop_converged_warning
9	classical_ml	100	0.9467	0.0533
1	closed_form_weighting	100	0.0295	0.5455
5	mi_closed_form	100	1.0000	0.0000
10	classical_ml	50	0.9674	0.0326
2	closed_form_weighting	50	0.0821	0.5234
6	mi_closed_form	50	1.0000	0.0000
11	classical_ml	20	0.9769	0.0224
3	closed_form_weighting	20	0.1838	0.4760
7	mi_closed_form	20	0.9973	0.0017
12	classical_ml	10	0.8850	0.0415
4	closed_form_weighting	10	0.2132	0.4668
8	mi_closed_form	10	0.9393	0.0342
	prop_converged_singular			
9			0.0000	
1			0.4250	
5			0.0000	
10			0.0000	
2			0.3945	
6			0.0000	
11			0.0007	
3			0.3402	
7			0.0010	
12			0.0735	
4			0.3200	
8			0.0265	



Coverage rate of β_3

	method	n	coverage95_beta3
9	classical_ml	100	0.9443000
1	closed_form_weighting	100	0.5914000
5	mi_closed_form	100	0.9339000
10	classical_ml	50	0.9355000
2	closed_form_weighting	50	0.6261000
6	mi_closed_form	50	0.9314000
11	classical_ml	20	0.9087000
3	closed_form_weighting	20	0.7038966
7	mi_closed_form	20	0.9174000
12	classical_ml	10	0.8407000
4	closed_form_weighting	10	0.7673192
8	mi_closed_form	10	0.8867000



Computation Time

	method	n	time_mean_seconds	time_median_seconds
9	classical_ml	100	0.028924	0.03
1	closed_form_weighting	100	11.017131	10.92
5	mi_closed_form	100	2.427138	2.34
10	classical_ml	50	0.022747	0.02
2	closed_form_weighting	50	3.128681	3.11
6	mi_closed_form	50	1.216079	1.18
11	classical_ml	20	0.019383	0.02
3	closed_form_weighting	20	0.771173	0.75
7	mi_closed_form	20	0.804032	0.77
12	classical_ml	10	0.018354	0.02
4	closed_form_weighting	10	0.331843	0.32
8	mi_closed_form	10	0.700160	0.68

Median computation time by Method and Sample Size

